

DUMBRAVITA DE CODRU, Romania

WMO: 151380

Lat: 46.6447N

Lon: 22.1711E

Elev: 1928

StdP: 13.70

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.3	17.7	4.6	7.5	20.8	9.7	9.7	22.9	26.5	41.6	18.9	36.0	3.1	160	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.8	87.8	68.8	84.4	68.1	81.7	66.9	72.0	82.5	70.3	80.7	68.7	79.0	3.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.9	114.0	77.0	66.9	106.2	75.9	65.2	99.9	74.6	37.1	82.6	35.5	81.0	34.1	79.4	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	14.0	11.2	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.8	31.2	36.4	41.8	51.5	60.2	66.8	69.2	71.4	61.6	52.8	44.5	34.1		
	DBStd	15.98	8.86	8.37	8.05	10.32	7.40	6.76	6.48	7.59	9.11	8.96	9.81	8.30		
	HDD50	2164	585	386	275	108	10	0	0	0	7	78	217	498		
	HDD65	5419	1049	802	719	411	185	59	31	27	173	388	616	959		
	CDD50	2840	1	4	21	154	326	503	594	663	355	163	52	4		
	CDD65	619	0	0	0	7	36	112	160	224	71	9	0	0		
	CDH74	3948	0	0	0	16	115	639	1048	1720	406	4	0	0		
	CDH80	1023	0	0	0	0	5	120	282	524	92	0	0	0		
(14) Wind	WSAvg	3.8	3.4	4.5	4.0	4.2	3.9	3.5	3.5	3.4	3.7	3.3	4.7	3.9		
(15) Precipitation	PrecAvg	34.1	2.5	1.9	2.2	2.8	3.1	4.1	3.8	2.6	3.1	2.6	2.6	2.8		
	PrecMax	43.7	3.9	5.0	7.4	4.8	6.5	8.7	7.2	4.7	7.4	5.9	4.7	6.3		
	PrecMin	23.2	1.1	0.4	0.6	0.8	1.3	1.1	1.4	0.5	0.6	0.1	0.2	0.4		
	PrecStd	5.6	0.8	1.0	1.5	1.1	1.3	1.6	1.6	1.3	1.6	1.5	1.0	1.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.1	59.0	66.8	75.8	80.3	87.0	91.0	92.5	87.3	74.2	68.2	56.6		
		MCWB	45.0	47.3	52.3	58.9	63.6	71.0	70.5	70.5	66.9	59.5	54.8	46.0		
	2%	DB	49.9	54.1	61.1	72.5	77.3	83.5	86.4	88.7	82.1	70.9	63.0	52.8		
		MCWB	43.7	46.0	49.3	56.4	63.5	69.6	69.3	68.6	64.1	57.5	52.6	44.6		
	5%	DB	46.5	51.1	57.4	69.1	74.6	80.8	82.8	85.5	78.2	68.2	60.6	48.7		
		MCWB	41.3	44.3	47.0	54.2	62.3	67.8	68.2	67.8	62.3	55.9	51.4	42.1		
	10%	DB	43.7	48.2	53.7	65.5	71.6	77.8	79.5	82.2	75.0	65.3	57.3	45.4		
		MCWB	39.6	42.1	44.9	52.5	60.8	66.6	66.5	66.7	61.3	55.1	49.1	40.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.5	49.9	53.8	61.1	67.8	73.9	74.2	75.1	69.3	63.1	56.1	47.7		
		MCDB	50.8	56.4	64.3	73.0	75.4	84.2	83.9	82.7	83.4	70.7	64.8	53.7		
	2%	WB	44.7	46.8	50.6	58.2	65.0	71.3	71.8	71.6	65.9	59.7	53.7	45.5		
		MCDB	49.0	51.8	60.6	69.2	74.2	81.0	81.3	83.5	77.4	66.7	62.2	52.0		
	5%	WB	42.7	44.9	47.8	55.9	63.0	69.5	70.1	69.8	63.9	57.8	52.2	43.1		
		MCDB	45.8	50.1	55.9	66.2	72.8	79.1	79.6	81.6	75.4	65.7	59.4	47.8		
	10%	WB	39.7	43.2	45.5	53.8	61.3	67.7	68.2	68.1	62.2	56.1	50.5	41.4		
		MCDB	43.0	47.4	52.6	64.1	70.6	76.9	77.2	80.0	72.9	64.0	56.6	45.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	10.6	12.9	14.7	14.1	14.5	15.6	14.8	12.9	11.2	9.4	8.3		
		MCDBR	11.2	13.4	16.4	17.0	16.1	16.8	18.0	16.3	15.6	13.5	11.6	10.4		
		MCWBR	7.4	8.2	9.0	9.3	10.5	9.7	9.7	8.1	7.6	7.8	6.6	6.4		
	5% WB	MCDBR	9.7	12.2	15.1	16.3	15.1	15.5	15.8	15.9	14.2	12.6	10.9	9.8		
		MCWBR	7.2	7.8	9.4	9.6	10.5	10.2	10.3	9.4	8.0	8.0	6.7	7.0		
(40) Clear-Sky Solar Irradiance	taub		0.305	0.329	0.365	0.413	0.399	0.398	0.412	0.414	0.385	0.353	0.328	0.306		
	taud		2.383	2.340	2.290	2.175	2.266	2.297	2.260	2.254	2.311	2.398	2.422	2.417		
	Ebn at Noon		249	265	271	267	275	275	269	263	261	253	237	235		
	Edh at Noon		23	29	36	44	41	40	41	40	35	27	22	20		
(44) All-Sky Solar Radiation	RadAvg		365	624	1004	1411	1708	1875	1840	1685	1189	802	456	284		
	RadStd		48	88	143	141	163	147	157	159	134	93	56	39		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (68 neighbors)	+1.26	N/A	N/A	N/A	+1.80	N/A	N/A	-229	-344	+202	+103

Nomenclature: See separate page